



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded

ISE-I (Semester-I) 2025-26

Class: First Year M. Tech.

Branch: All branches

Course: Open Elective: Operations Research

Course Code: OEC-803

Date: 17-Oct-2025

Time: 16:30 to 17:15

Max. Marks: 20

Q. No.	Questions	Marks	CO	BT
Q. 1)	Define systems engineering. Differentiate between systems approach and system analysis. How is it related to operations research?	07	1	1
Q. 2)	What is a model? Explain in brief role of systems models and state the different types of system models.	07	3	2
Q. 3)	Explain the concept of system with a neat diagram.	06	4	2

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S. G. G. S. Institute of Engineering and Technology, Vishnupuri, Nanded
Department of Production Engineering
ISE-I (Sem-I) Academic Year: 2025-26

Class: First Year M. Tech (CAD/CAM)	Subject: New Product Design	Subject Code: PCC-DM509
Date: 16/10/2025	Time: 16:30 TO 17:15	Max. Marks: 20

Note

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Figures to right indicate full marks.
- iii. The symbols carry usual meaning.
- iv. Mobile phones and programmable calculators are strictly prohibited.
- v. Exchange/Sharing of stationery, calculator etc. not allowed.

Q. N.		M	CO	BT
Q.1	Describe the physical realizability.	05	CO1	BT1
Q.2	Explain the morphology of design process.	05	CO2	BT3
Q.3-A	Explain the essential factors of product design.	10	CO1	BT2
	OR			
Q.3-B	Describe the principles of good product design according to customer and manufacturer perspective.	10	CO2	BT1

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Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

- CO1. Understand the concept & importance of new product design.
- CO2. Study the steps, requirements & specification of product design.
- CO3. Understand the selection of material & manufacturing processes in design.
- CO4. Understand the design for various purpose & cost estimation.
- CO5. Study product development approaches & design project management.